Chidaksh Ravuru

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Research Interests

Building rigorous evaluation frameworks that create tight feedback loops between data, evals, and model behavior: experimental design, statistical analysis, and LLM evaluation methodologies for reasoning and alignment, alongside causal verification of what a model actually relies on. Extending this same diagnostic rigor to red-teaming and adversarial evaluation, characterizing failure modes in safety-relevant model behavior before high-stakes releases.

Education

2024–2026 **University of North Carolina at Chapel Hill**, MS in Computer Science
Advisor: Prof. Shashank Srivastava, GPA: 4.0/4.0
Relevant Courses: Reinforcement Learning (RLHF/PPO), Reliable ML, Advanced LLM Agents

2020–2024 **Indian Institute of Technology, Dharwad**, B.Tech in Computer Science
Advisors: Prof. Prabuchandran K.J. & Prof. Rajshekhar Bhat, GPA: 9.24/10

Publications

2026 **UNMASK: Discovering and Causally Verifying Spurious Shortcuts in Classifiers**, *COLM 2026*
Chidaksh Ravuru*, Shashank Srivastava

2026 **Can LLMs Understand What We Cannot Say? Measuring Multilevel Alignment Through Abortion Stigma**, *Preprint*
Anika Sharma, Malavika Mampally, **Chidaksh Ravuru***, Kandyce Brennan, Neil Gaikwad

2024 **Agentic Retrieval-Augmented Generation for Time Series Analysis**, *KDD 2024*
Chidaksh Ravuru*, Sagar Sakhinana, Venkataramana Runkana

2024 **RESTORE: Graph Embedding Assessment Through Reconstruction**, *Preprint*
Hong Yung Yip, **Chidaksh Ravuru***, Neelabha Banerjee, Shashwat Jha, Amit Sheth, Aman Chadha, Amitava Das

Workshop Papers

2024 **Parameter-Efficient Quantized Mixture-of-Experts Meets Vision-Language Instruction Tuning**, *ICML, Workshop on Foundational Models*
Sagar Sakhinana, **Chidaksh Ravuru***, Geethan Sannidhi, Venkataramana Runkana

Research Experience

Jun 2026–Present **AI Engineer**, *Excipy LLC*, Remote
Architected and built an end-to-end agentic evaluation system for healthcare AI research.
- Built an **end-to-end agentic system** (Python, LangChain/LangGraph) that autonomously researches drugs, excipients, and market dynamics, validating its own outputs against **structured rubrics** and **systematically flagging failure modes** before reaching a user-facing product.
- Designed and deployed **retrieval and validation pipelines** with structured tool use to extract, cross-check, and verify domain-specific data from unstructured sources, with **human-in-the-loop review** gates ensuring output reliability.

Summer 2025 **Applied Scientist Intern**, *Pavo AI*
Trained persona-conditioned Process Reward Models (PRMs) to guide search-space exploration in a deployed planning agent.
- Built **PersonaBench**, a synthetic dataset of **3,000 positive/negative trajectories** using a **student-teacher framework**: student extracts company profiles from the web, teacher verifies sources and flags hallucinations, then LLMs generate persona-conditioned reasoning traces reflecting each company's ML maturity and infrastructure constraints.
- Trained **persona-conditioned PRMs** on PersonaBench, conditioning on both **correctness and persona fit** to select responses suited to each company's capabilities, improving task performance by **10%** in production; reward model outputs additionally used to flag **behavioral failure modes** in agent responses before deployment.

Summer 2023 **Machine Learning Research Intern**, *Tata Research, Development and Design Centre*
Designed Agentic RAG based orchestration system for time series analysis.
- Developed an agentic RAG pipeline with structured **data curation**, **DPO**-based post-training, and multi-step retrieval orchestration, achieving a **12% improvement** in prediction accuracy across temporal forecasting tasks, contributing to the **KDD 2024 publication**.
- Designed an **automated evaluation pipeline** across **diverse task benchmarks**, measuring model capability, identifying failure modes, and driving **18% performance gains** through targeted fine-tuning, establishing a feedback loop between evaluation findings and model improvement.

Projects

2026 **LLM-as-Judge Evaluation with Statistical Rigor**, *Code* | *Dashboard*

- Evaluated frontier LLMs as automated judges across 5 behavioral dimensions, implementing **bootstrap confidence intervals**, **Cohen's d** effect sizes, and **Krippendorff's alpha** for inter-rater reliability and **position bias detection**.
- Applied **experimental design** via causal minimal-pair testing to isolate single-dimension effects on output quality, validating rubric sensitivity with dimension-weighted scoring and close-call detection via paired bootstrap testing.

2026 **UNMASK: Discovering and Causally Verifying Spurious Shortcuts in Classifiers**, *COLM 2026*

- Built a fully automated evaluation pipeline that generates candidate spurious features, filters via a multi-stage statistical protocol with **FDR control**, and establishes causal model dependence through counterfactual interventions, achieving **81%** causal reliance reduction and **+12.7 pp** adversarial robustness on HANS.
- Generalized to RewardBench2 reward model annotations, surfacing spurious correlations such as prompt-response lexical overlap rewarding parroting over substantive quality, without task-specific modification.

2026 **Unlearning Robustness: Signal×Attack Diagnostic Matrix**, *LessWrong Blog*

- Designed a diagnostic framework evaluating whether LLM safety interventions genuinely remove target knowledge or leave **deceptively intact residuals**, computing attack-free static signals on unlearned checkpoints and testing whether each signal predicts vulnerability to its matched **red-teaming attack**.
- Powered the validation to **32 checkpoints** (8 methods × 4 hyperparameter variants) via structured attacks (quantization, relearning, direction ablation), finding weight-space distance significantly predicts attack recovery ($\rho = -0.85$, **p<0.001**, BH-FDR $q < 0.05$), while confirming no single static signal alone certifies safety.

2026 **Does Structure Survive Scale? Diagnosing Hierarchy in SAEs**, *EleutherAI SOAR, Ongoing*

- Collaborating within an **EleutherAI research pod** of PhD students, advised by **Gonçalo Paulo**, to build a coverage-based diagnostic suite quantitatively evaluating whether hierarchy-recovery methods (Matryoshka SAEs, Temporal SAEs, Temporal Feature Analysis) produce coherent parent-child feature structures on real LLMs (Gemma-2-2B).
- Designing controlled PCFG experiments with tunable distributional properties to isolate which properties of natural language cause hierarchy-recovery methods to fail, and training SAEs on PCFG-trained transformer activations to benchmark against TinyStories and Gemma Scope SAEs.

Skills

Libraries: PyTorch, HuggingFace Transformers, PEFT/LoRA, TransformerLens, OpenUnlearning, LangChain, Scikit-learn

Methods: LLM-as-Judge Evaluation, Evaluation Frameworks, Experimental Design, Statistical Analysis (Bootstrap CIs, Effect Sizes), Benchmark Design, Causal Verification, Adversarial Evaluation & Red-Teaming, Interpretability, RLHF/RLAIF, DPO

Programming: Python, C++, Bash/Shell Script, SQL, JAX, LaTeX

Infrastructure: Docker, AWS, Weights & Biases, Git, Slurm, Unix/Linux, vLLM

Teaching & Achievements

Summer 2025 **Summer School Teacher**, *UNC School of Medicine*
Led a lecture series for high school students on Statistics and Machine Learning (Linear Regression, SVMs) in neuroscience and medical imagery.

2025–2026 **Graduate Teaching Assistant**, *UNC School of Data Science and Society*
TA for DATA 110: Intro to Data Science. Conducted lab recitations and taught basic probability, statistics, and Python coding for undergraduates.

Reviewing **EMNLP, ICLR, COLM, AIES**

Awards **BlueDot Impact**: AI Safety Technical Course, Frontier AI Governance Course · **EleutherAI SOAR Fellow**, advised by **Gonçalo Paulo** · Top 1% for **Google Research Week** (India, 2024) · Fully funded at **MLSS 2022** (Krakow) · **4th place** Inter IIT Techfest 2023